

Condition (ii). Set of vectors
 $V = \{V_1, V_2, \dots, V_n\}$ must be linearly independent

$$C_1 V_1 + C_2 V_2 + C_3 V_3 + \dots + C_n V_n = 0$$

$$\Rightarrow C_1(1, 0, 0) + C_2(0, 1, 0) + C_3(0, 0, 1) = (0, 0, 0)$$

$$\Rightarrow (C_1, C_2, C_3) = (0, 0, 0)$$

$$\Rightarrow C_1 = 0$$

$$\Rightarrow C_2 = 0$$

$$\Rightarrow C_3 = 0$$

We get the trivial solution $C_1 = 0, C_2 = 0$
& $C_3 = 0$. So, the set of vectors is linearly independent. As both conditions are satisfied hence set of vectors is the basis for $\mathbb{R}^3$.

NULLITY:

A matrix that satisfies the condition
Nullity of matrix = (number of unknown)
- Rank of matrix.

Q: Find the nullity of matrix:

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & 12 \end{bmatrix}$$

Using elementary row operation (echelon form)

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & -1 & -3 & 1 & 3 \\ 0 & 3 & 9 & 0 & 12 \end{bmatrix} \begin{array}{l} \\ R_2 \times (-1) \\ R_3 + 2R_1 \\ \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 21 \end{bmatrix} \begin{array}{l} \\ \\ R_3 + R_2 \\ R_4 - 3R_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 3 & 21 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Swapping R_3 by R_4

$\Rightarrow$ Rank of $A = 3$

$$\text{Nullity} = 5 - 3 = 2$$

L-3

Example: If $f_1(x) = \sin x$, $f_2(x) = \sin x + \cos x$, $f_3(x) = \sin x - \cos x$ be three functions, check whether the functions are linearly dependent or linearly independent, using Wronskian matrix.

$$f_1(x) = \sin x$$

iff w.r.t 'x'

$$f_1'(x) = \cos x$$

iff w.r.t 'x'

$$f_1''(x) = -\sin x$$

$$f_2(x) = \sin x + \cos x$$

$$f_2'(x) = \cos x - \sin x$$

$$f_2''(x) = -\sin x - \cos x$$

$$f_3(x) = \sin x - \cos x$$

$$f_3'(x) = \cos x + \sin x$$

$$f_3''(x) = -\sin x + \cos x$$

Applying Wronskian matrix.

$$W(x) = \begin{bmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{bmatrix} = \begin{bmatrix} \sin x & \sin x + \cos x & \sin x - \cos x \\ \cos x & \cos x - \sin x & \cos x + \sin x \\ -\sin x & -\sin x - \cos x & -\sin x + \cos x \end{bmatrix}$$

$$\Rightarrow \sin x \begin{bmatrix} \cos x - \sin x & \cos x + \sin x \\ -\sin x - \cos x & -\sin x + \cos x \end{bmatrix}$$

$$-(\sin x + \cos x) \begin{bmatrix} \cos x & \cos x + \sin x \\ -\sin x & -\sin x + \cos x \end{bmatrix}$$

$$+(\sin x - \cos x) \begin{bmatrix} \cos x & \cos x - \sin x \\ -\sin x & -\sin x - \cos x \end{bmatrix}$$

$$w(x) = \sin x [(-\sin x \cos x + \cos^2 x + \sin^2 x - \sin x \cos x) - (-\sin x \cos x - \sin^2 x - \cos^2 x - \sin x \cos x) - (\sin x + \cos x)(-\sin x \cos x + \cos^2 x) - (-\sin x \cos x - \sin^2 x) + (\sin x - \cos x)(-\sin x \cos x - \cos^2 x) - (-\sin x \cos x + \sin^2 x)]$$

$$= \sin^2 \theta + \cos^2 \theta \cdot 1$$

$$\Rightarrow \sin x (-2 \sin x \cos x + 1 + 2 \sin x \cos x + 1) - (\sin x + \cos x)(-\sin x \cos x + \cos^2 x + \sin x \cos x + \sin^2 x) + (\sin x - \cos x)(-\sin x \cos x - \cos^2 x + \sin x \cos x - \sin^2 x)$$

$$\Rightarrow 2 \sin x - (\sin x + \cos x) + (\sin x - \cos x)(-1)$$

$$\Rightarrow 2 \sin x - \sin x - \cos x - \sin x + \cos x$$

$$\Rightarrow 2 \sin x - 2 \sin x$$

$\Rightarrow$ Since $w(x) = 0$, so the given functions are linearly dependent.

INNER PRODUCT SPACES:

Let u, v & w be vectors in vector space V & let c be any scalar. An inner product on V is a function that associates with real numbers $\langle u, v \rangle$ with each pair of vectors.

$$U = (u_1, u_2, \dots, u_n), V = (v_1, v_2, \dots, v_n)$$

and satisfies the following axioms.

- 1) $\langle u, v \rangle = \langle v, u \rangle$
- 2) $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$
- 3) $c \langle u, v \rangle = \langle cu, v \rangle$
- 4) $\langle u, u \rangle \geq 0, \langle u, u \rangle = 0$ if & only if $u = 0$.

Note: $u \cdot v =$ dot product (Euclidean inner product for $\mathbb{R}^n$)

$\langle u, v \rangle =$ general inner product in vector space.

Q. Show that the dot product in $\mathbb{R}^n$ satisfies the four axioms of an inner product.

let,

$$u = (x_1, x_2, \dots, x_n)$$

$$v = (y_1, y_2, \dots, y_n)$$

$$w = (z_1, z_2, \dots, z_n)$$

1.

$$\langle u, v \rangle = u \cdot v = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$$\langle v, u \rangle = v \cdot u = y_1 x_1 + y_2 x_2 + \dots + y_n x_n$$

$$\langle u, v \rangle = \langle v, u \rangle$$

2.

$$\begin{aligned} \langle u, v+w \rangle &= x_1(y_1+z_1) + x_2(y_2+z_2) + \dots + x_n(y_n+z_n) \\ &= x_1 y_1 + x_1 z_1 + x_2 y_2 + x_2 z_2 + \dots + x_n y_n + x_n z_n \\ &= x_1 y_1 + x_2 y_2 + \dots + x_n y_n + x_1 z_1 + x_2 z_2 + \dots + x_n z_n \\ &= \langle u, v \rangle + \langle u, w \rangle \end{aligned}$$

$$\begin{aligned} 3. \quad \langle cu, v \rangle &= c(x_1 y_1 + x_2 y_2 + \dots + x_n y_n) \\ &= c x_1 y_1 + c x_2 y_2 + \dots + c x_n y_n \\ &= p_1 y_1 + p_2 y_2 + \dots + p_n y_n \end{aligned}$$

$$\begin{aligned} \text{let, } p_1 &= c x_1, p_2 = c x_2, p_n = c x_n \\ &= \langle p_1, p_2, \dots, p_n \rangle \langle y_1, y_2, \dots, y_n \rangle \\ &= \langle (c x_1, c x_2, \dots, c x_n), (y_1, y_2, \dots, y_n) \rangle \\ &= \langle c(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \rangle \\ &= \langle cu, v \rangle \end{aligned}$$

$$\begin{aligned} 4. \quad \langle u, u \rangle &= (x_1)(x_1) + (x_2)(x_2) + \dots + (x_n)(x_n) \\ &= x_1^2 + x_2^2 + \dots + x_n^2 \geq 0 \end{aligned}$$

$$\langle u, u \rangle = 0 \text{ if \& only if } u = (0, 0, \dots, 0)$$

Thus an inner product in $\mathbb{R}^n$. By Definition dot product satisfies all conditions.

Q. Show that the function defines an inner product in $\mathbb{R}^2$, where $u = (u_1, u_2)$,
 $v = (v_1, v_2)$ & $\langle u, v \rangle = u_1 v_1 + 2u_2 v_2$.

1.

$$\langle u, v \rangle = u_1 v_1 + 2u_2 v_2 = v_1 u_1 + 2v_2 u_2 = \langle v, u \rangle$$

$$\begin{aligned} 2. \langle u, v+w \rangle &= \langle (u_1, u_2), (v_1+v_2, w_1+w_2) \rangle \\ &= (u_1+v_1)w_1 + 2(u_2+v_2)w_2 \\ &= u_1 w_1 + v_1 w_1 + 2u_2 w_2 + 2v_2 w_2 \\ &= u_1 w_1 + 2u_2 w_2 + v_1 w_1 + 2v_2 w_2 \\ &= \langle u, w \rangle + \langle v, w \rangle. \end{aligned}$$

$$3. c\langle u, v \rangle = c(u_1 v_1 + 2u_2 v_2) = cu_1 v_1 + 2cu_2 v_2.$$

$$\text{let, } z_1 = cu_1, z_2 = cu_2.$$

$$= z_1 v_1 + 2z_2 v_2.$$

$$= \langle z, v \rangle$$

$$= \langle (cu_1, cu_2), v \rangle = \langle c(u_1, u_2), v \rangle = c\langle u, v \rangle$$

$$4. \langle u, u \rangle = \langle (u_1, u_2), (u_1, u_2) \rangle = u_1^2 + 2u_2^2 \geq 0$$

Inner product satisfied.

Q. Show that the following function is not an inner product in $\mathbb{R}^3$
 $\langle u, v \rangle = u_1 v_1 - 2u_2 v_2 + u_3 v_3$.

$$\text{let, } u = (1, 2, 1)$$

$$\begin{aligned} \langle u, u \rangle &= u_1 u_1 - 2u_2 u_2 + u_3 u_3 \\ &= u_1^2 - 2u_2^2 + u_3^2 \end{aligned}$$

$$\begin{aligned} \langle u, u \rangle &= (1)^2 - 2(2)^2 + (1)^2 \\ &= 1 - 8 + 1 = -6 < 0 \end{aligned}$$

Thus, the given function is not inner product in $\mathbb{R}^3$.

NORM (LENGTH) OF u :

$$\|u\| = \sqrt{\langle u, u \rangle}$$

distance b/w two vectors.

$$d(u, v) = \sqrt{\langle u-v, u-v \rangle} = \|u-v\|$$

Angle b/w two vectors.

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}, \quad 0 \leq \theta \leq \pi$$

Obtuse angle $\langle u, v \rangle < 0$

Acute angle $\langle u, v \rangle > 0$

Right angle $\langle u, v \rangle = 0$

Orthogonal ($u \perp v$)

u & v are orthogonal if $\langle u, v \rangle = 0$

If $\|v\| = 1$, thus v is called unit vector

Q. let $p(x) = 1 - 2x^2$, $q(x) = 4 - 2x + x^2$ be polynomial in P_2 .

a) $\langle p, q \rangle = ?$

b) $\|q\| = ?$

c) $d(p, q) = ?$

$$\begin{aligned} \text{a) } \langle p, q \rangle &= (1)(4) + (0)(-2) + (-2)(1) \\ &= 4 - 0 - 2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{b) } \|q\| &= \sqrt{(4)^2 + (-2)^2 + (1)^2} \\ &= \sqrt{16 + 4 + 1} \\ &= \sqrt{21} \end{aligned}$$

$$\begin{aligned} \text{c) } d(p, q) &= \|p - q\| \\ &= \|p - q\| = \|(1 - 2x^2) - (4 - 2x + x^2)\| \\ &= \|-3 + 2x - 3x^2\| \\ &= \sqrt{(-3)^2 + (2)^2 + (-3)^2} \\ &= \sqrt{9 + 4 + 9} \\ &= \sqrt{22} \end{aligned}$$

GRAM SCHMIDT ORTHO NORMALIZATION PROCESS

(L-4)

Let $B = \{u_1, u_2, \dots, u_n\}$ is a basis for an inner product space V .

Step 1: let $v_1 = u_1$

Step 2: $v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$

Step 3: $v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$

$v_n = u_n - \frac{\langle u_n, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_n, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle u_n, v_3 \rangle}{\|v_3\|^2} v_3 \dots$

$B' = \{v_1, v_2, \dots, v_n\}$ is an Orthogonal basis

$B'' = \left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \frac{v_3}{\|v_3\|}, \dots, \frac{v_n}{\|v_n\|} \right\}$ is an orthonormal basis

2. Apply Gram Schmidt process to the following basis $B = \{(1, 1, 0), (1, 2, 0), (0, 1, 2)\}$
 $\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}$
 Find the orthonormal basis.

Let $u_1 = (1, 1, 0)$ $u_2 = (1, 2, 0)$ $u_3 = (0, 1, 2)$

STEP: 1

let $v_1 = u_1$ $v_1 = (1, 1, 0)$

$\therefore v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$

$$\langle u_2, v_1 \rangle = 3$$

$$\|v_1\|^2 = 2$$

$$\Rightarrow v_2 = (1, 2, 0) - \frac{3}{2} (1, 1, 0)$$

$$\Rightarrow v_2 = (1 - \frac{3}{2}, 2 - \frac{3}{2}, 0 - 0) = (-\frac{1}{2}, \frac{1}{2}, 0)$$

$\therefore v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$

$$\langle u_3, v_1 \rangle = 1$$

$$\langle u_3, v_2 \rangle = \frac{1}{2}$$

$$\|v_1\|^2 = 2$$

$$\|v_2\|^2 = (-\frac{1}{2})^2 + (\frac{1}{2})^2 + 0^2 = \frac{1}{4} + \frac{1}{4} \Rightarrow (\frac{1}{2})$$

$$\Rightarrow V_3 = (0, 1, 2) - \frac{1}{2}(1, 1, 0) - \frac{1/2}{1/2}(-1/2, 1/2, 0)$$

$$\Rightarrow V_3 = \left(0 - \frac{1}{2} + \frac{1}{2}, 1 - \frac{1}{2} - \frac{1}{2}, 2 - 0 - 0 \right)$$

$$\Rightarrow V_3 = (0, 0, 2)$$

$$B' = \{(1, 1, 0), (-1/2, 1/2, 0), (0, 0, 2)\}$$

is an orthogonal basis.

$$: W_1 = \frac{V_1}{\|V_1\|} \Rightarrow \frac{(1, 1, 0)}{\sqrt{2}} \Rightarrow \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$: W_2 = \frac{V_2}{\|V_2\|} \Rightarrow \frac{(-1/2, 1/2, 0)}{\sqrt{2}} \Rightarrow \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right)$$

$$: W_3 = \frac{V_3}{\|V_3\|} \Rightarrow \frac{(0, 0, 2)}{2} \Rightarrow (0, 0, 1)$$

$$B'' = \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right), (0, 0, 1) \right\}$$

is an orthonormal vector.

2. Find the coordinates of $w = (5, -5, 2)$ relative to the following orthonormal basis of $\mathbb{R}^3$

$$B = \left\{ \left(\frac{3}{5}, \frac{4}{5}, 0 \right), \left(-\frac{4}{5}, \frac{3}{5}, 0 \right), (0, 0, 1) \right\}$$

$$\begin{aligned} \langle w, v_1 \rangle &= 5 \times \frac{3}{5} + (-5) \times \frac{4}{5} + 2 \times 0 \\ &= 3 - 4 + 0 \Rightarrow -1 \end{aligned}$$

$$\begin{aligned} \langle w, v_2 \rangle &= 5 \times \left(-\frac{4}{5} \right) + (-5) \times \left(\frac{3}{5} \right) + 2 \times 0 \\ &= -4 - 3 + 0 \Rightarrow -7 \end{aligned}$$

$$\begin{aligned} \langle w, v_3 \rangle &= 5 \times 0 + (-5) \times 0 + 2 \times 1 \\ &= 2 \end{aligned}$$

$$[w]_B = (-1, -7, 2)$$

EIGEN VALUE & EIGEN VECTOR:

Let A be an $n \times n$ matrix with real components. The number λ (real or complex) is called an eigen value of A if there is a non-zero vector V in $\mathbb{C}^n$ such that

$$AV = \lambda V$$

The vector $V \neq 0$ is called an eigen-vector of A corresponding to the eigen value λ .

Example:

$$A = \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix} \Rightarrow A - \lambda I = \begin{bmatrix} 2 & -18 \\ 6 & -11 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 20 & -18 \\ 12 & -11 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow \text{eigen vector}$$

eigen value

THEOREM: Let A be an $n \times n$ matrix. Then λ is an eigenvalue of A if & only if $P(\lambda) = |A - \lambda I| = 0$

Procedure for computing Eigenvalue & eigenvectors.

- i) Find $P(\lambda) = |A - \lambda I|$
- ii) Find the roots $\lambda_1, \lambda_2, \dots, \lambda_m$ of $P(\lambda)$
- iii) Corresponding to each eigen value λ_i , solve the homogeneous system $(A - \lambda_i I)V = 0$.

$$A = \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix}$$

STEP: 1

$$P(\lambda) = |A - \lambda I|$$

$$P(\lambda) = \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$P(\lambda) = \begin{bmatrix} 10-\lambda & -18 \\ 6 & -11-\lambda \end{bmatrix}$$

STEP: 2

$$P(\lambda) = \begin{bmatrix} 10-\lambda & -18 \\ 6 & -11-\lambda \end{bmatrix} = 0$$

$$(10-\lambda)(-11-\lambda) - (-18)(6) = 0$$

$$-110 - 10\lambda + 11\lambda + \lambda^2 + 108 = 0$$

$$\lambda^2 + \lambda - 2 = 0$$

$$\lambda^2 + 2\lambda - \lambda - 2 = 0$$

$$\lambda(\lambda+2) - 1(\lambda+2) = 0$$

$$(\lambda+2)(\lambda-1) = 0$$

$$\lambda_2 = -2, \lambda_1 = 1$$

STEP: 3

Eigen vector for $\lambda_1 = 1$

$$(A - \lambda_1 I)V = 0$$

$$\Rightarrow \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 9 & -18 \\ 6 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & -18 \\ 6 & -12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \begin{matrix} R_1 \div 9 \\ R_2 \div 6 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} R_2 - R_1$$

$$\Rightarrow x_1 - 2x_2 = 0$$

$$\Rightarrow x_1 = 2x_2$$

$$\Rightarrow x_1 = 2 \text{ let } x_2 = 1$$

$$V_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

For $\lambda = -2$

$$(A - \lambda I)$$

$$(A + 2I)V = 0$$

$$\Rightarrow \begin{bmatrix} 10 & -18 \\ 6 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 12 & -18 \\ 6 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_1 \rightarrow \begin{bmatrix} 12 & -18 \\ 6 & -9 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3/2 \\ 6 & -9 \end{bmatrix} R_1 \div 12$$

$$R_2 \rightarrow \begin{bmatrix} 1 & -3/2 \\ 0 & 0 \end{bmatrix} R_2 - 6R_1$$

$$\Rightarrow x_1 - \frac{3}{2}x_2 = 0$$

$$\Rightarrow x_1 = \frac{3}{2}x_2 \quad \text{let } x_2 = 2$$

$$\Rightarrow x_1 = 3$$

$$V_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

let $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ Find the eigen value & eigen vector.

$$\therefore |A - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & -1 & 4 \\ 3 & 2 - \lambda & -1 \\ 2 & 1 & -1 - \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 - \lambda & -1 & 4 \\ 3 & 2 - \lambda & -1 \\ 2 & 1 & -1 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda) \begin{vmatrix} 2 - \lambda & -1 \\ 1 & -1 - \lambda \end{vmatrix} - (-1) \begin{vmatrix} 3 & -1 \\ 2 & -1 - \lambda \end{vmatrix} + 4 \begin{vmatrix} 3 & 2 - \lambda \\ 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda)(-2 - \lambda + \lambda^2 + 1) + (-3 - 3\lambda + 2) + 4(3 - 4 + 2\lambda) = 0$$

$$\Rightarrow (1 - \lambda)(\lambda^2 - \lambda - 1) + (-1 - 3\lambda) + 4(2\lambda - 1) = 0$$

$$\Rightarrow \lambda^2 - \lambda - 1 - \lambda^2 + \lambda^2 + \lambda - 1 - 3\lambda + 8\lambda - 4 = 0$$

$$\Rightarrow -\lambda^3 + 2\lambda^2 + 5\lambda - 6 = 0$$

$$\Rightarrow \lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0$$

$$\therefore \lambda = 1$$

$$\begin{array}{r|rrrr} 1 & 1 & -2 & -5 & 6 \\ & & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

$$\lambda^2 - \lambda - 6 = 0$$

$$\lambda^2 - 3\lambda + 2\lambda - 6 = 0$$

$$\lambda(\lambda - 3) + 2(\lambda - 3) = 0$$

$$(\lambda + 2)(\lambda - 3) = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = -2, \lambda_3 = 3$$

$\Rightarrow$ Eigen value vector for $\lambda_1 = 1$

$$(A - \lambda_1 I)V = 0$$

$$\Rightarrow \left\{ \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 0 & -1 & 4 \\ 3 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$= A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & -1 & 4 \\ 2 & 1 & -2 \end{bmatrix} \text{ swap } R_1 \text{ with } R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 2 & -1 & -2 \end{bmatrix} \begin{matrix} R_1 - R_3 \\ (-1) R_2 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & -4 \\ 0 & -1 & 4 \end{bmatrix} \begin{matrix} \\ R_2 - 3R_1 \\ R_3 - 2R_1 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_2 \times -2 \\ R_3 - R_2 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1/2 & -2 \\ 0 & 1 & -10 \\ 0 & 0 & -6 \end{bmatrix} \begin{matrix} \\ \\ R_3 + R_2 \end{matrix}$$

$$V_1 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$$

$$\Rightarrow x_1 + x_3 = 0 \text{ when}$$

$$\Rightarrow x_2 - 4x_3 = 0 \text{ when}$$

$$x_2 = 4x_3$$

$$\therefore \text{let } x_3 = 1$$

$$\Rightarrow x_2 = 4$$

$$\Rightarrow x_1 + 1 = 0$$

$$\Rightarrow x_1 = -1$$

For $\lambda_2 = -2$
 $(A - \lambda_2 I)V = 0$
 $(A + 2I)V = 0$

$$\begin{bmatrix} 3 & -1 & 4 \\ 3 & 4 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$B = \begin{bmatrix} 1 & -2 & 3 \\ 3 & 4 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{matrix} R_1 - R_3 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 10 & -10 \\ 0 & 5 & -5 \end{bmatrix} \begin{matrix} R_2 - 3R_1 \\ R_3 - 2R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{matrix} R_2 \div 10 \\ R_3 \div 5 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2$$

$$x_1 - 2x_2 + 3x_3 = 0$$

$$x_2 - x_3 = 0$$

$$x_2 = x_3$$

$$\therefore x_3 = 1$$

$$x_2 = 1$$

$$x_1 - 2 + 3 = 0$$

$$x_1 = -1$$

$$V_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

For $\lambda_3 = 3$

$$(A - 3I)V = 0$$

$$\begin{bmatrix} -2 & -1 & 4 \\ 3 & -1 & -1 \\ 2 & 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2 & -1 & 4 \\ 1 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_2 + R_1 \\ R_3 + R_1 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 3 \\ -2 & -1 & 4 \\ 0 & 0 & 0 \end{bmatrix} \text{ swap } R_1 \text{ \& } R_2$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 3 \\ 0 & -5 & 10 \\ 0 & 0 & 0 \end{bmatrix} R_2 + 2R_1$$

$$x_1 - 2x_2 + 3x_3 = 0$$

$$-5x_2 + 10x_3 = 0$$

$$x_2 = 2x_3$$

$$= x_3 = 1$$

$$x_2 = 2$$

$$x_1 - 2(2) + 3(1) = 0$$

$$x_1 - 4 + 3 = 0$$

$$x_1 = 1$$

$$V_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{Trace} = a_{11} + a_{22} + a_{33} \dots a_{nn}$$

PROPERTIES OF EIGENVALUES

1. Any square matrix A & its transpose have the same eigen value.
2. Sum of eigen values of a matrix is equal to the trace of matrix, where trace is known as the sum of elements of the principle diagonal.
3. The product of eigen values of matrix A is equal to the determinant of A .
4. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigen values of a matrix A , then the eigen values of
 - (i) kA are $k\lambda_1, k\lambda_2, \dots, k\lambda_n$.
 - (ii) A^m are $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$.
 - (iii) A^{-1} are $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$.

DIAGONALIZATION OF A MATRIX

Theorem: If a square matrix A of order n has linearly independent eigen vectors, then a matrix P can be found such that PAP^{-1} is a diagonal matrix.

$$\therefore D = P^{-1}AP$$

Q1. let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ Find the diagonal matrix.

$$\therefore |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (6-\lambda) \begin{vmatrix} 3-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 2 & 3-\lambda \end{vmatrix} + 2 \begin{vmatrix} -2 & 3-\lambda \\ 2 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (6-\lambda) [(3-\lambda)(3-\lambda) - (-1)(-1)] + 2 [(2)(3-\lambda) - (-2)(2)] + 2 [(-2)(-1) - (3-\lambda)(2)] = 0$$

$$\Rightarrow (6-\lambda)(9-3\lambda-3\lambda+\lambda^2-1) + 2(18-6\lambda-3\lambda+\lambda^2) + 2(2-6+2\lambda) = 0$$

$$\Rightarrow (6-\lambda)(8-6\lambda+\lambda^2) + 2(18-9\lambda+\lambda^2) + 2(-4+2\lambda) = 0$$

$$48 - 36\lambda + 6\lambda^2 - 8\lambda + 6\lambda^2 - \lambda^3 + 36 - 18\lambda - 8 + 4\lambda = 0$$

$$\Rightarrow -\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

$$\Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\Rightarrow \lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 8$$

Eigen vector $\lambda_1 = 2 \quad \therefore (A - \lambda_1 I)V = 0$

$$\Rightarrow \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$B \Rightarrow \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad 2R_3 - R_1$$

$$\Rightarrow \begin{bmatrix} 1 & -1/2 & 1/2 \\ -2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_1 \div 4$$

$$\Rightarrow \begin{bmatrix} 1 & -1/2 & 1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 + 2R_1$$

$$\Rightarrow x_1 - \frac{1}{2}x_2 + \frac{1}{2}x_3 = 0$$

$$\Rightarrow \text{let } x_2 \text{ \& } x_3 = 1$$

$$x_1 - \frac{1}{2} + \frac{1}{2} = 0$$

$$\Rightarrow x_1 = 0$$

$$\Rightarrow V_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{let } x_2 = 3, x_3 = 1$$

$$\Rightarrow x_1 - \frac{1}{2}(3) + \frac{1}{2}(1) = 0$$

$$\Rightarrow x_1 - 1 = 0 \Rightarrow x_1 = 1$$

Eigen vector for $\lambda = 8 : (A - \lambda I)V = 0$

$$\Rightarrow \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow C = \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}$$

$$R_2 - R_1$$

$$R_3 + R_1$$

$$\Rightarrow \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 + R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -1 \\ 0 & +1 & +1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 \div - \\ R_2 \div -3 \\ \end{matrix}$$

$$\Rightarrow \text{let } x_1 + x_2 - x_3 = 0 \dots (i)$$

$$\Rightarrow x_2 + x_3 = 0 \dots (ii)$$

$$\Rightarrow x_2 = -x_3 \quad \text{let } x_3 = 1$$

$$\Rightarrow x_2 = -1$$

$$\Rightarrow (i) \Rightarrow x_1 - 1 - 1 = 0$$

$$\Rightarrow x_1 = 2$$

$$V_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$|P| = 0 - 1(1+1) + 2(1-3) = -2 - 4 = -6$$

$$= -1(2) + 2(-2)$$

$$= -2 - 4$$

$$= -6$$

(linearly dependent)

Inverse find ho sakta ha q k $-6 \neq 0$ ha.

Finding P^{-1} by Gauss-Jordan

$$[P, I] = P^{-1}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 3 & -1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \text{change } R_1 \text{ with } R_2$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -2 & 2 & -1 & -1 & 1 \end{array} \right] R_3 - R_1$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 6 & 2 & -1 & 1 \end{array} \right] R_3 + 2R_2$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/3 & -1/6 & 1/6 \end{array} \right] R_3 \div 6$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -7/3 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1/3 & 1/3 & -2/3 \\ 0 & 0 & 1 & 1/3 & -1/6 & 1/6 \end{array} \right] \begin{array}{l} R_1 - 3R_2 \\ R_2 - 2R_3 \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2/3 & -1/6 & 7/6 \\ 0 & 1 & 0 & 1/3 & 1/3 & -2/3 \\ 0 & 0 & 1 & 1/3 & -1/6 & 1/6 \end{array} \right] R_1 + 7R_3$$

$$\rightarrow P^{-1} = \begin{bmatrix} -2/3 & -1/6 & 7/6 \\ 1/3 & 1/3 & -1/3 \\ 1/3 & -1/6 & 1/6 \end{bmatrix}$$

$$\therefore D = P^{-1} A P$$

$$\Rightarrow D = \begin{bmatrix} -2/3 & -1/6 & 7/6 \\ 1/3 & 1/3 & -1/3 \\ 1/3 & -1/6 & 1/6 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2/3 & -1/6 & 7/6 \\ 1/3 & 1/3 & -1/3 \\ 1/3 & -1/6 & 1/6 \end{bmatrix} \begin{bmatrix} 0 & 2 & 16 \\ 2 & 6 & -8 \\ 2 & 2 & 8 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 - 2/6 + 14/6 & -4/3 - 6/6 + 14/6 & -32/3 + 8/6 + 56/6 \\ 0 + 2/3 - 2/3 & 2/3 + 6/3 - 2/3 & 16/3 - 8/3 - 8/6 \\ 0 - 2/6 + 2/6 & 2/3 - 6/6 + 2/6 & 16/3 + 8/6 + 8/6 \end{bmatrix}$$

$$\Rightarrow D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

CALEY-HAMILTON THEOREM:

Statement: Every square matrix satisfies its own characteristic equation.

$$\text{Home work} = \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$A^3 - 12A^2 + 36A - 32I = 0$$

$$\text{where } A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Thus if A is a square matrix & if $\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$ --- (i) be its characteristic equation then according to Caley-Hamilton Theorem (i) it is satisfied.

$$A^3 - 12A^2 + 36A - 32I = 0 \text{ --- (ii)}$$

we can find A^{-1} we get.

$$A^3 - 12A^2 + 36A - 32I = 0$$

$$A^{-1}A^3 - 12A^{-1}A^2 + 36A^{-1}A - 32A^{-1}I = (A^{-1})(0)$$

$$A^2 - 12A + 36I - 32A^{-1} = 0$$

$$A^2 - 12A + 36I = 32A^{-1} \therefore A^{-1}A = I$$

$$A^2 - 12A + 36I = 32A^{-1}$$

$$A^{-1} = \frac{1}{32}[A^2 - 12A + 36I]$$

Q. Find the characteristic equation of the given matrix & hence find the A^{-1}

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

Characteristic equation.

$$\therefore |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & -2 \\ 1 & 1-\lambda & 1 \\ 1 & 3 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \begin{vmatrix} 1-\lambda & 1 \\ 3 & -1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ 1 & -1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 1-\lambda \\ 1 & 3 \end{vmatrix}$$

$$\Rightarrow (1-\lambda)[(1-\lambda)(-1-\lambda) - (1)(3)] - 2[(1)(-1-\lambda) - (1)(1)] - 2[(1)(3) - (1-\lambda)(1)] = 0$$

$$\Rightarrow (1-\lambda)(-1-\lambda+\lambda^2-3) - 2(-1-\lambda-1) - 2(3-1+\lambda) = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2-4) - 2(-\lambda-2) - 2(\lambda+2) = 0$$

$$\Rightarrow \lambda^2-4-\lambda^3+\lambda^2+2\lambda+4-2\lambda-4 = 0$$

$$\Rightarrow -\lambda^3+\lambda^2+4\lambda-4 = 0$$

$$\Rightarrow \lambda^3-\lambda^2-4\lambda+4 = 0$$

By applying Cayley-Hamilton theorem.

$$\Rightarrow A^3 - A^2 - 4A + 4I = 0$$

Pre multiply by A^{-1}

$$\Rightarrow A^2 - A - 4I + 4A^{-1} = 0$$

$$\Rightarrow 4A^{-1} = -A^2 + A + 4I \quad \dots (a)$$

$$\Rightarrow A^{-1} = \frac{1}{4}[-A^2 + A + 4I]$$

For (a)

$$A^2 = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 2 \\ 3 & 6 & -2 \\ 3 & 2 & 2 \end{bmatrix}$$

$$\Rightarrow 4A^{-1} = \begin{bmatrix} -1+2-2 \\ -3-6+2 \\ -3-2-2 \end{bmatrix} + \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\Rightarrow 4A^{-1} = \begin{bmatrix} 4 & 4 & -4 \\ -2 & -1 & 3 \\ -2 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 1/4 & 1/4 & -1/4 \\ -1/2 & -1/4 & 3/4 \\ -1/2 & 1/4 & 1/4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1/2 & -1/4 & 3/4 \\ -1/2 & 1/4 & 1/4 \end{bmatrix}$$

QR FACTORIZATION :

The QR factorization decomposes any matrix into the product of orthonormal matrix & upper triangular matrix mathematically.

$$A = QR$$

where A is any matrix, Q is orthonormal matrix & R is upper triangular matrix.

PROCEDURE:

Step 1: Compute orthonormal matrix $Q = [u_1 \ u_2 \ u_3]$ by using Gram-Schmidt.

Step 2: Compute upper triangular matrix by using inner product.

$$R = \begin{bmatrix} \langle u_1, u_1 \rangle & \langle u_2, u_1 \rangle & \langle u_3, u_1 \rangle \\ 0 & \langle u_2, u_2 \rangle & \langle u_3, u_2 \rangle \\ 0 & 0 & \langle u_3, u_3 \rangle \end{bmatrix}$$

Q. Find the QR-decomposition of matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

let $u_1 = (1, 1, 0)$, $u_2 = (1, 0, 1)$, $u_3 = (0, 1, 1)$

Applying Gram-Schmidt process find the orthogonal & orthonormal vector.

$$v_1 = u_1$$

$$\Rightarrow v_1 = (1, 1, 0)$$

$$\therefore v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$\langle u_2, v_1 \rangle = 1$$

$$\|v_1\|^2 = \sqrt{2}$$

$$\Rightarrow v_2 = (1, 0, 1) - \frac{1}{\sqrt{2}} (1, 1, 0)$$

$$\Rightarrow v_2 = (1 - \frac{1}{\sqrt{2}}, 0 - \frac{1}{\sqrt{2}}, 1 - 0)$$

$$\Rightarrow v_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1 \right)$$

$$V_3 = U_3 - \frac{\langle U_3, V_1 \rangle}{\|V_1\|^2} V_1 - \frac{\langle U_3, V_2 \rangle}{\|V_2\|^2} V_2$$

$$\langle U_3, V_1 \rangle = 1$$

$$\langle U_3, V_2 \rangle = 1/2$$

$$\|V_2\|^2 = \sqrt{1/4 + 1/4 + 1} = \sqrt{6}/2 \Rightarrow 3/2$$

$$V_3 = (0, 1, 1) - \frac{1}{2} (1, 1, 0) - \frac{1/2}{3/2} \left(\frac{1}{2}, -\frac{1}{2}, 1 \right)$$

$$\Rightarrow V_3 = (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0 \right) - \frac{1}{3} \left(\frac{1}{2}, -\frac{1}{2}, 1 \right)$$

$$\Rightarrow V_3 = \left(0 - \frac{1}{2} - \frac{1}{6}, 1 - \frac{1}{2} + \frac{1}{6}, 1 - 0 - \frac{1}{3} \right)$$

$$\Rightarrow V_3 = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right) \text{ orthogonal vectors}$$

Now finding orthonormal vectors.

$$W_1 = \frac{V_1}{\|V_1\|} = \frac{(1, 1, 0)}{\sqrt{2}} \Rightarrow \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$W_2 = \frac{V_2}{\|V_2\|} = \frac{(1/2, -1/2, 1)}{\sqrt{6}/2} \Rightarrow \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)$$

$$W_3 = \frac{V_3}{\|V_3\|} = \frac{(-2/3, 2/3, 2/3)}{\sqrt{12}/3} \Rightarrow \left(-\frac{2}{\sqrt{12}}, \frac{2}{\sqrt{12}}, \frac{2}{\sqrt{12}} \right)$$

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

$$R = \begin{bmatrix} \langle U_1, W_1 \rangle & \langle U_2, W_1 \rangle & \langle U_3, W_1 \rangle \\ 0 & \langle U_2, W_2 \rangle & \langle U_3, W_2 \rangle \\ 0 & 0 & \langle U_3, W_3 \rangle \end{bmatrix}$$

$$\langle U_1, W_1 \rangle = 1/\sqrt{2} + 1/\sqrt{2} \Rightarrow 2/\sqrt{2}$$

$$\langle U_2, W_1 \rangle = 1/\sqrt{2}$$

$$\langle U_3, W_1 \rangle = 1/\sqrt{2}$$

$$\langle U_2, W_2 \rangle = 1/\sqrt{6} + 2/\sqrt{6} \Rightarrow 3/\sqrt{6}$$

$$\langle U_3, W_2 \rangle = -1/\sqrt{6} + 2/\sqrt{6} \Rightarrow 1/\sqrt{6}$$

$$\langle U_3, W_3 \rangle = 1/\sqrt{3} + 1/\sqrt{3} \Rightarrow 2/\sqrt{3}$$

$$R = \begin{bmatrix} 2/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 3/\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

HOUSE HOLDER FACTORIZATION / LEAST SQUARE METHOD

Step 1. QR factorization of coefficient matrix.

Step 2. Find the solution of system of equation by using $Rx = Q^T b$

This method is used to find the solution of system of linear equation either is square or rectangular.

Ex: Find the solution of following system of equation.

$$x + 3y + 5z = 3$$

$$x + y + 0z = 5$$

$$x + y + 2z = 7$$

$$x + 3y + 3z = -3$$

By using House Holder factorization.

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

let $u_1 = (1, 1, 1, 1)$, $u_2 = (3, 1, 1, 3)$
 $u_3 = (5, 0, 2, 3)$

$\therefore V_1 = u_1$
 $\Rightarrow V_1 = (1, 1, 1, 1)$

$\therefore V_2 = u_2 - \frac{\langle u_2, V_1 \rangle}{\|V_1\|^2} V_1$

$\langle u_2, V_1 \rangle = 8$
 $\|V_1\|^2 = 4$

$\Rightarrow V_2 = (3, 1, 1, 3) - \frac{8}{4} (1, 1, 1, 1)$

$\Rightarrow V_2 = (3-2, 1-2, 1-2, 3-2)$

$\Rightarrow V_2 = (1, -1, -1, 1)$

$\therefore V_3 = u_3 - \frac{\langle u_3, V_1 \rangle}{\|V_1\|^2} V_1 - \frac{\langle u_3, V_2 \rangle}{\|V_2\|^2} V_2$

$\langle u_3, V_1 \rangle = 10$

$\langle u_3, V_2 \rangle = 6$

$\|V_1\|^2 = 4$

$\|V_2\|^2 = 4$

$\Rightarrow V_3 = (5, 0, 2, 3) - \frac{10}{4} (1, 1, 1, 1) - \frac{3}{4} (1, -1, -1, 1)$

$\Rightarrow V_3 = (5, 0, 2, 3) - \left(\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, \frac{5}{2}\right) - \left(\frac{3}{2}, -\frac{3}{2}, -\frac{3}{2}, \frac{3}{2}\right)$

$\Rightarrow V_3 = \left(\frac{5-5-3}{2}, \frac{0-5+3}{2}, \frac{2-5+3}{2}, \frac{3-5-3}{2}\right)$

$\Rightarrow V_3 = (1, -1, 1, -1)$

$\therefore W_1 = \frac{V_1}{\|V_1\|} \Rightarrow \frac{(1, 1, 1, 1)}{2} \Rightarrow \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

$\therefore W_2 = \frac{V_2}{\|V_2\|} \Rightarrow \frac{(1, -1, -1, 1)}{2} \Rightarrow \left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

$\therefore W_3 = \frac{V_3}{\|V_3\|} \Rightarrow \frac{(1, -1, 1, -1)}{2} \Rightarrow \left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$

$Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

$$R = \begin{bmatrix} \langle u_1, w_1 \rangle & \langle u_2, w_1 \rangle & \langle u_3, w_1 \rangle \\ 0 & \langle u_2, w_2 \rangle & \langle u_3, w_2 \rangle \\ 0 & 0 & \langle u_3, w_3 \rangle \end{bmatrix}$$

$$\langle u_1, w_1 \rangle = 2$$

$$\langle u_2, w_1 \rangle = 4$$

$$\langle u_3, w_1 \rangle = 5$$

$$\langle u_2, w_2 \rangle = 2$$

$$\langle u_3, w_2 \rangle = 3$$

$$\langle u_3, w_3 \rangle = 2$$

$$R = \begin{bmatrix} 2 & 4 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

$$Rx = Q^T b$$

$$\Rightarrow Q^T b = \begin{bmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \\ -3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3/2 + 5/2 + 1/2 - 3/2 \\ 3/2 - 5/2 - 1/2 - 3/2 \\ 3/2 - 5/2 + 1/2 + 3/2 \end{bmatrix} \Rightarrow \begin{bmatrix} 6 \\ -6 \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \\ 4 \end{bmatrix}$$

$$\Rightarrow 2x + 4y + 5z = 6 \quad \text{--- (i)}$$

$$\Rightarrow 2y + 3z = -6 \quad \text{--- (ii)}$$

$$\Rightarrow 2z = 4 \Rightarrow \boxed{z = 2}$$

$$(ii) \Rightarrow 2y + 3(2) = -6$$

$$\Rightarrow 2y + 6 = -6$$

$$\Rightarrow y = -12/2 \Rightarrow \boxed{y = -6}$$

$$(i) \Rightarrow 2x + 4(-6) + 5(2) = 6$$

$$\Rightarrow 2x - 14 = 6$$

$$\Rightarrow x = 20/2 \Rightarrow \boxed{x = 10}$$

$$\begin{aligned} Q. \quad & x_1 - 2x_2 - 7x_3 + 7x_4 = 5 \\ & -x_1 + 2x_2 + 8x_3 - 5x_4 = -7 \\ & 3x_1 - 4x_2 - 17x_3 + 13x_4 = 14 \\ & 2x_1 - 2x_2 - 11x_3 + 8x_4 = 7 \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 \\ -1 & 2 & 8 & -5 \\ 3 & -4 & -17 & 13 \\ 2 & -2 & -11 & 8 \end{bmatrix} \begin{bmatrix} 5 \\ -7 \\ 14 \\ 7 \end{bmatrix}$$

$$\text{let } u_1 = v_1$$

$$\Rightarrow u_1 = (1, -1, 3, 2)$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$\Rightarrow \langle u_2, v_1 \rangle = -20$$

$$\Rightarrow \|v_1\|^2 = 15$$

$$v_2 = (-2, 2, -4, -2) - (-20/15)(1, -1, 3, 2)$$

$$v_2 = (-2, 2, -4, -2) + (4/3, -4/3, 4, 8/3)$$

$$v_2 = (-2/3, 2/3, 0, 2/3)$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$\Rightarrow \langle u_3, v_1 \rangle = -28$$

$$\Rightarrow \langle u_3, v_2 \rangle = 8/3$$

$$\Rightarrow \|v_1\|^2 = 15$$

$$\Rightarrow \|v_2\|^2 = 2/9$$

$$\cancel{v_3 = (-2/3, 2/3, 0, 2/3)}$$

$$\begin{aligned} \Rightarrow v_3 &= (-7, 8, -11, -11) + \frac{28}{15}(1, -1, 3, 2) \\ &\quad - \frac{8/3}{2/9}(-2/3, 2/3, 0, 2/3) \end{aligned}$$

$$\Rightarrow v_3 = (3/15, 12/15, 3/5, 91/15) \Rightarrow (1/5, 4/5, 3/5, 31/5)$$

$$v_4 = u_4 - \frac{\langle u_4, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_4, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle u_4, v_3 \rangle}{\|v_3\|^2} v_3$$

(L-6)

LINEAR TRANSFORMATIONS:

Let U & V be two vector spaces over the same field & let $T: U \rightarrow V$ be a function. The T is called a linear transformation. If the following conditions are satisfied.

- (i) $T(u_1 + u_2) = T(u_1) + T(u_2)$, $u_1, u_2 \in U$
 (ii) $T(au) = aT(u)$, $a \in F$ & for all $u \in U$
 or $\mathbb{R}$

$$T(au_1 + bu_2) = aT(u_1) + bT(u_2).$$

Example: Check $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by
 $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$.

It is linear or not?

$$\text{let } u_1 = (x_1, x_2, x_3)$$

$$u_2 = (y_1, y_2, y_3)$$

$$a, b \in \mathbb{R}$$

$$au_1 + bu_2 = a(x_1, x_2, x_3) + b(y_1, y_2, y_3) \\ = (ax_1, ax_2, ax_3) + (by_1, by_2, by_3)$$

$$T(ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) \\ = (ax_1 + by_1 - ax_2 - by_2, ax_1 + by_1 - ax_3 - by_3) \\ = (ax_1 - ax_2 + by_1 - by_2, ax_1 - ax_3 + by_1 - by_3) \\ = (ax_1 - ax_2, ax_1 - ax_3) + (by_1 - by_2, by_1 - by_3) \\ = a(x_1 - x_2, x_1 - x_3) + b(y_1 - y_2, y_1 - y_3) \\ = aT(x_1, x_2, x_3) + bT(y_1, y_2, y_3) \\ = aT(u_1) + bT(u_2) = T(au_1 + bu_2).$$

Hence T is a linear transformation.

Check $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by
 $T(x_1, x_2, x_3) = (x_1 + 1, x_2 + x_3)$.
 Is it linear or not.

let $u_1 = (x_1, x_2, x_3)$ $u_2 = (y_1, y_2, y_3)$
 $a, b \in \mathbb{R}$

$$au_1 + bu_2 = a(x_1, x_2, x_3) + b(y_1, y_2, y_3)$$

$$= (ax_1, ax_2, ax_3) + (by_1, by_2, by_3)$$

$$T(ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) = (ax_1 + by_1 + 1, ax_2 + by_2 + ax_3 + by_3)$$

$$= (ax_1 + 1 + by_1, ax_2 + ax_3 + by_2 + by_3)$$

$$= (ax_1 + 1/a, x_2 + x_3) + b(y_1, y_2 + y_3)$$

$$T(au_1 + bu_2) \neq aT(u_1) + bT(u_2)$$

Hence T is not a linear transformation.

let linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$
 be given by $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$
 write in matrix form with standard basis.

let $u_1 = (1, 0, 0)$ $u_2 = (0, 1, 0)$ $u_3 = (0, 0, 1)$ in $\mathbb{R}^3$
 $v_1 = (1, 0)$ $v_2 = (0, 1)$ in $\mathbb{R}^2$

$$T(u_1) = av_1 + bv_2$$

$$T(u_1) = T(1, 0, 0) = (1, 1) = a(1, 0) + b(0, 1)$$

$$= (a, 0) + (0, b)$$

$$= (a + 0, 0 + b)$$

$$(1, 1) = (a, b)$$

$$a = 1, b = 0$$

$$T(u_1) = 1 \cdot v_1 + 0 \cdot v_2$$

$$T(u_2) = T(0, 1, 0) = (-1, 0) = av_1 + bv_2$$

$$(-1, 0) = (a, b)$$

$$T(u_2) = -1 \cdot v_1 + 0 \cdot v_2$$

$$T(u_3) = T(0, 0, 1) = (0, -1) = av_1 + bv_2$$

$$(0, -1) = (a, b)$$

$$T(u_3) = 0 \cdot v_1 - 1 \cdot v_2$$

Check $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by
 $T(x_1, x_2, x_3) = (x_1 + 1, x_2 + x_3)$
 Is it linear or not.

let $u_1 = (x_1, x_2, x_3)$ $u_2 = (y_1, y_2, y_3)$
 $a, b \in \mathbb{R}$

$$au_1 + bu_2 = a(x_1, x_2, x_3) + b(y_1, y_2, y_3)$$

$$= (ax_1, ax_2, ax_3) + (by_1, by_2, by_3)$$

$$T(ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) = (ax_1 + by_1 + 1, ax_2 + by_2 + ax_3 + by_3)$$

$$= (ax_1 + 1 + by_1, ax_2 + ax_3 + by_2 + by_3)$$

$$= a(x_1 + 1/a, x_2 + x_3) + b(y_1, y_2 + y_3)$$

$$T(au_1 + bu_2) \neq aT(u_1) + bT(u_2)$$

Hence T is not a linear transformation.

let linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$
 be given by $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$
 write in matrix form with standard basis.

let $u_1 = (1, 0, 0)$ $u_2 = (0, 1, 0)$ $u_3 = (0, 0, 1)$ in $\mathbb{R}^3$
 $v_1 = (1, 0)$ $v_2 = (0, 1)$ in $\mathbb{R}^2$

$$T(u_1) = av_1 + bv_2$$

$$T(u_1) = T(1, 0, 0) = (1, 1) = a(1, 0) + b(0, 1)$$

$$= (a, 0) + (0, b)$$

$$= (a + 0, 0 + b)$$

$$(1, 1) = (a, b)$$

$$a = 1, b = 0$$

$$T(u_1) = 1 \cdot v_1 + 0 \cdot v_2$$

$$T(u_2) = T(0, 1, 0) = (-1, 0) = av_1 + bv_2$$

$$(-1, 0) = (a, b)$$

$$T(u_2) = -1 \cdot v_1 + 0 \cdot v_2$$

$$T(u_3) = T(0, 0, 1) = (0, -1) = av_1 + bv_2$$

$$(0, -1) = (a, b)$$

$$T(u_3) = 0 \cdot v_1 - 1 \cdot v_2$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

The matrix of A with respect to standard basis

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be a linear transformation defined by

$$T(x_1, x_2, x_3) = (x_2 + x_3, x_1 + x_2, x_1, x_1 - x_2)$$

Find the matrix of T with respect to the basis $B = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$ & the basis $E = \{w_1, w_2, w_3, w_4\}$ for $\mathbb{R}^4$ where

$$v_1 = (1, 1, 0), v_2 = (1, 0, -1), v_3 = (0, 1, 0)$$

$$w_1 = (1, -1, 0, 0), w_2 = (1, 0, 1, 0), w_3 = (0, 1, 0, 0),$$

$$w_4 = (0, 0, 1, 1).$$

From Definition

$$T(v_1) = a_1 w_1 + a_2 w_2 + a_3 w_3 + a_4 w_4$$

$$T(v_1) = T(1, 1, 0) = (1, 2, 1, 0)$$

$$= a_1(1, -1, 0, 0) + a_2(1, 0, 1, 0) + a_3(0, 1, 0, 0) + a_4(0, 0, 1, 1)$$

$$= (a_1 + a_2, -a_1 + a_3, a_2 + a_4, a_4)$$

$$a_1 + a_2 = 1 \Rightarrow a_1 = 0$$

$$-a_1 + a_3 = 2 \Rightarrow a_3 = 2$$

$$a_2 + a_4 = 1 \Rightarrow a_2 = 1$$

$$a_4 = 0$$

$$T(v_1) = 0 \cdot w_1 + 1 \cdot w_2 + 2 \cdot w_3 + 0 \cdot w_4$$

$$T(v_2) = a_1 w_1 + a_2 w_2 + a_3 w_3 + a_4 w_4$$

$$T(1, 0, -1) = (-1, 1, 1, 1)$$

$$= (a_1 + a_2, -a_1 + a_3, a_2 + a_4, a_4)$$

$$a_1 + a_2 = -1 \Rightarrow a_1 = -1$$

$$-a_1 + a_3 = 1 \Rightarrow a_3 = 0$$

$$a_2 + a_4 = 1 \Rightarrow a_2 = 0$$

$$a_4 = 1$$

$$T(v_2) = -1 \cdot w_1 + 0 \cdot w_2 + 0 \cdot w_3 + 1 \cdot w_4$$

$$T(v_3) = a_1 w_1 + a_2 w_2 + a_3 w_3 + a_4 w_4$$

$$T(0, 1, 0) = (1, 1, 0, -1)$$

$$= (a_1 + a_2, -a_1 + a_3, a_2 + a_4, a_4)$$

$$a_1 + a_2 = 1 \Rightarrow a_1 = 0$$

$$-a_1 + a_3 = 1 \Rightarrow a_3 = 1$$

$$a_2 + a_4 = 0 \Rightarrow a_2 = -1$$

$$a_4 = -1$$

$$T(v_3) = 0 \cdot w_1 + 1 \cdot w_2 + 1 \cdot w_3 + (-1) \cdot w_4$$

The matrix A of T with respect to the given basis is

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 7 & 3 \\ 3 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + 7x_2 + 3x_3 \\ 3x_1 + 4x_2 + 9x_3 \end{bmatrix}$$

The matrix of a linear transformation

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

Determine T in terms of coordinates of a vector $x \in \mathbb{R}^3$

let $x = (x_1, x_2, x_3)$. T is given by

$$T(x) = Ax = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 3x_1 + x_2 + 2x_3 \\ x_2 + x_3 \\ -x_1 + x_2 + x_3 \end{bmatrix}$$

$$T(x) = (3x_1 + x_2 + 2x_3, x_2 + x_3, -x_1 + x_2 + x_3)$$

Power se if square matrix ki find ho sakte ha.

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \text{ find } A^*$$

$$\therefore |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 1-\lambda \begin{vmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 2-\lambda \\ 2 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 1-\lambda [(2-\lambda)(3-\lambda) - (1)(2)] - 1 [(2)(1) - (2)(2-\lambda)] = 0$$

$$\Rightarrow 1-\lambda (6-2\lambda-3\lambda+\lambda^2-2) - 1(2-4+4\lambda) = 0$$

$$\Rightarrow 1-\lambda (\lambda^2-5\lambda+4) - 1(4\lambda-2) = 0$$

$$\Rightarrow \lambda^2-5\lambda+4-\lambda^3+5\lambda^2-4\lambda-4\lambda+2 = 0$$

$$\Rightarrow -\lambda^3+6\lambda^2-13\lambda+6 = 0$$

$$\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$$

$$\text{For } \lambda_1 \quad (A - \lambda_1 I) = 0$$

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ 2 & 2 & 2 \end{bmatrix} \text{ change } R_1 \text{ by } R_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - 2R_1$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ -x_3 &= 0 \Rightarrow x_3 = 0 \end{aligned}$$

$$x_1 + x_2 = 0$$

$$\Rightarrow x_1 = -x_2 \quad \text{let } x_2 = 1$$

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{For } \lambda_2 = 2 \quad (A - \lambda_2 I) = 0$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 2 & -1 \end{bmatrix} \begin{array}{l} R_2 + R_1 \\ R_3 + 2R_1 \end{array}$$

$$\begin{aligned} -x_1 - x_3 &= 0 \\ 2x_2 - x_3 &= 0 \\ x_2 &= x_3/2 \end{aligned}$$

$$\text{let } x_3 = 2$$

$$x_2 = 1$$

$$x_1 = -2$$

$$v_2 = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{For } \lambda_3 = 3$$

$$(A - \lambda_3 I) = 0$$

$$\Rightarrow \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ -2 & 0 & -1 \\ 2 & 2 & 0 \end{bmatrix} \begin{array}{l} R_1 \text{ by } R_2 \\ R_3 + 2R_1 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 1 \\ 0 & 4 & -2 \end{bmatrix} \begin{array}{l} R_2 + 2R_1 \\ R_3 - 2R_1 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 + 2R_2 \end{array}$$

$$x_1 - x_2 + x_3 = 0$$

$$-2x_2 + x_3 = 0$$

$$x_2 = x_3/2$$

$$x_3 = 2$$

$$x_2 = 1$$

$$x_1 = -1$$

$$Y_3 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$[P, I] \rightarrow [I, P^{-1}]$$

$$\rightarrow \begin{bmatrix} -1 & -2 & -1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ -1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_1 \times (-1) \end{array}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1/2 \end{bmatrix} \quad \begin{array}{l} R_2 + R_1 \\ R_3 \div 2 \end{array}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1/2 \end{bmatrix} \quad R_3 - R_2$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1/2 \end{bmatrix} \quad R_1 - R_2$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & -1/2 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1/2 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ 1 & 1 & 1/2 \end{bmatrix}$$

$$D = P^{-1} A P$$

$$A^+ = P D^+ P^{-1}$$

$$= \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1^+ & 0 & 0 \\ 0 & 2^+ & 0 \\ 0 & 0 & 3^+ \end{bmatrix} \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ 1 & 1 & 1/2 \end{bmatrix}$$